Flame Freefall

Make NASA combustion evidence usable for the next research decision.

Find relevant experiments. Audit the comparison. Export the evidence and its limits.

Prevention

Material flammability

Detection

Smoke and sensors

Suppression

Droplet extinction

Code Nashai / Muhammad Marjan Ahmed / Solo builder

The hard part is knowing what transfers.

A spacecraft-fire researcher asks:

Which experiments inform a study near a proposed Lunar atmosphere?

Find

Tables and reports are spread across investigations.

Compare

Matching oxygen alone does not match the experiment.

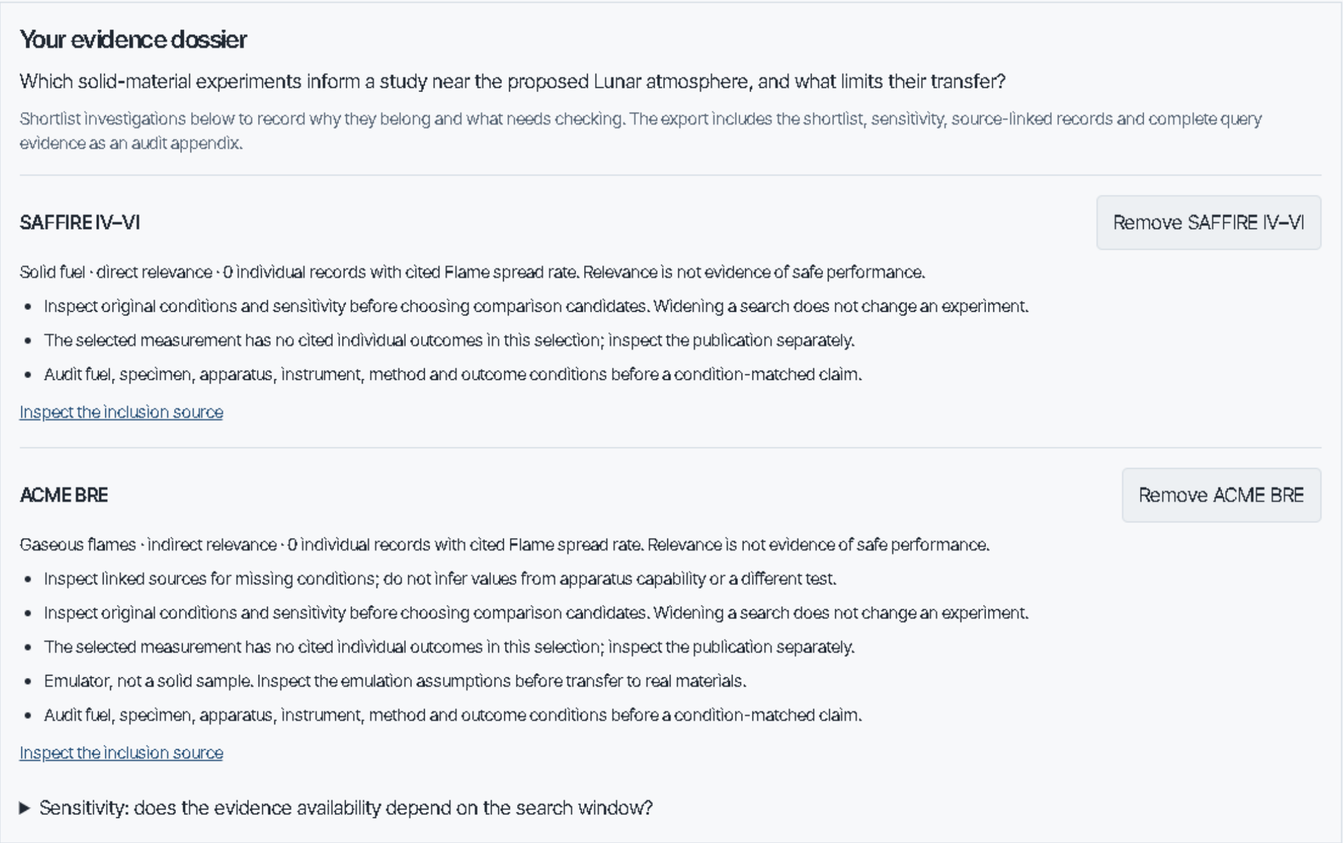
Interpret

A nearby result can still have limited applicability.

The opportunity: preserve context from the research question to the source.

Source: supplied challenge-details.md; NASA PSI investigations and governed source reports

One question. A traceable evidence dossier.



Actual local dossier capture, cropped from the saved workflow run; not a live deployment check.

1 Set the question

Choose job and conditions.

2 Inspect the evidence

Check relevance and source limits.

3 Keep the reasoning

Export settings, records and citations.

Implemented: system/site/app/decide; lib/decision.ts; lib/psi-comparison.ts; docs/evidence-method.md

Demo: an atmosphere match is a starting point.

Research question

What informs a solid-material study near a proposed Lunar atmosphere?

SAFFIRE IV–VI

Direct solid-fuel evidence.
Inspect setup and missing fields.

ACME BRE

Indirect condensed-fuel emulation.
Keep its physical geometry distinct.

Widening the oxygen search window changes the retrieved evidence.

The experiment has not changed. Neither has its transfer limit.

Two temperatures. No invented conclusion.

77°C

VI-3 / two-sided PMMA

57°C

VI-4 / one-sided PMMA

Reported observations from Table 2

Audit the comparison

Fuel description differs.
Oxygen and pressure differ.
Method and dimensions remain incomplete.

Inspect both results. Do not infer a controlled effect.

AI helps you ask. Code preserves the evidence.

1,951
Indexed records

1,838
Test / test-point records

23
Stable governed documents

Built to inspect

Source-linked normalization
Python / TypeScript query parity
Deterministic measurement widgets

Boundaries stay visible

Missing metadata and source conflicts
Unknown observation independence
No material-safety or prediction claim

Make the next research decision defensible.

For researchers and aerospace engineers

A source-backed shortlist.
An auditable comparison.
A reproducible starting point.

Next: prove the benefit

Observe researcher task completion.
Review relevance with domain experts.
Resolve critical source metadata.

Try the evidence workflow

/decide/ → shortlist → audit → export

Project: github.com/marjan-ahmed/flame-freefall

Designed to support research; measured time savings and safety outcomes are not established.